Ghilmo Choi

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Education

Korea Aerospace University, Republic of Korea — Mar 2021 – Feb 2027 (expected)
B.S. in Mechanical Engineering

- Official GPA: **4.00 / 4.5** (Major GPA: **4.08 / 4.5**)
- Expected GPA after current-semester grades are posted: **4.08 / 4.5** (Major GPA: **4.18 / 4.5**)
- Main Coursework: Introduction to Machine Learning, Automatic Control, Mechanical Vibrations, Data Structures and Algorithms, Probabilistic Robotics

Research Interests

- On-device vision-language models for natural-language robotic mission execution
- Perception, Prediction, and Decision Making for Autonomous Driving
- AI model architectures and efficient computation for robotic intelligence

Research / Project Experience

ASOL (Autonomous Systems and Optimization Lab) — Korea Aerospace Univ.
- *Undergraduate Researcher* — June 2025 – Present

- **On-device VLM-based robotic mission execution**
  Developed a natural-language mission execution framework that converts user commands into executable robot behaviors on a UGV platform
- **Hallucination-aware and SLAM-grounded task planning**
  Designed an Actor–Guard–Critic planning loop and integrated RTAB-Map, object detection, structured world states, and ROS2/Nav2 control for real-robot language-to-motion execution
- **AirSim-based perception, reconstruction, and path generation**
  Extending the framework with YOLOE, NanoOWL, and Grounding DINO for object detection on AirSim data, NVBlox for 3D reconstruction, and OMPL for path generation

KAUVOY Autonomous Driving Club — Korea Aerospace Univ.
- *Localization Part* — Nov 2024 – Nov 2025

- **EKF-based localization**
  Implemented Dual EKF localization on an ERP42 autonomous vehicle by fusing IMU, wheel encoder, and RTK-GPS measurements
- **Autonomous parking and path tracking**
  Developed and tested a real-vehicle parking pipeline covering cone perception, parking-space decision, Hybrid A* path planning, and Pure Pursuit-based path tracking

- *Team Leader* — Nov 2025 – present

- **Finite-State-Machine-based decision-making module for autonomous driving scenarios**

Course Project, Introduction to Machine Learning — Korea Aerospace Univ.
- **End-to-End Machine Learning Classification and Regression Pipeline** — Aug 2025 – Nov 2025

- Built and validated classification and regression pipelines using preprocessing, dimensionality reduction, ensemble methods, and FCNNs; achieved **0.915 macro F1** on a 4-class imbalanced classification task

Honors & Activities

- Finalist, **2025 College Mobility Innovation Competition**
- Excellence Award, **Drone X AI Marketing Challenge** (2026)
- 2026 - **Airbus 101 Competition** (5th Place)

Publications

- **Co-author**, "*On-Device Vision Language Model Based Natural Language Mission Execution Framework*," submitted to the International Conference on Control, Automation and Information Sciences (**ICCAIS**), 2025
- **Co-author**, "*ROS `robot_localization` Dual EKF: Verification of Localization Accuracy and Application to Autonomous Driving Path Tracking*," submitted to the Korean Society of Automotive Engineers (**KSAE**), 2025

Technical Skills

- Python, C/C++, MATLAB, Linux(Ubuntu), ROS1, ROS2, Docker, Git